

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as the dominant paradigm for advancing the reasoning capabilities of large language models. The prevailing family of algorithms exemplified by GRPO, however, is systematically undermined by vanishing gradients—a failure mode we trace to two coupled and mutually reinforcing design choices: the fixed clipping bounds imposed on token-level probability ratios, and the step-local standardization of identical rewards. Acting in concert, these choices produce ineffective gradient updates and leave a large fraction of generated responses entirely unexploited. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a unified policy-optimization method that resolves both bottlenecks at their source through two complementary and jointly designed mechanisms. First, a dynamic-adaptive clipping strategy widens the admissible update interval for each token as a monotone function of its prior probability, thereby restoring token-level exploration precisely where it is most informative. Second, a smooth advantage standardization procedure aggregates reward statistics across cumulative training steps, substantially improving the response-level utilization of generated data. Acting together, these mechanisms establish DCPO as a state-of-the-art method across four mathematical reasoning benchmarks and four distinct models spanning both instruct and base variants. On the AIME24 benchmark with Qwen2.5-Math-7B, DCPO attains an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-fold sampling, decisively surpassing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9); on the AIME25 benchmark with Qwen2.5-14B, DCPO reaches (23.3/19.0), again outperforming GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). These accuracy gains are inseparable from concomitant efficiency gains: DCPO raises the nonzero-advantage ratio by an average of 28% over GRPO across the four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude relative to both GRPO and DAPO, all while delivering superior end-task performance. Collectively, these results demonstrate that DCPO converts generated data into usable learning signal markedly more efficiently, furnishing a general and effective foundation for reinforcement learning in large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus of contemporary research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Among the techniques driving this progress, Reinforcement Learning from Verifiable Rewards (RLVR) has established itself as the dominant paradigm: by optimizing LLMs against rule-based outcome rewards, RLVR delivers its most pronounced gains precisely in those domains that demand reflective, multi-step decision-making, most notably mathematics and coding (DeepSeek-AI et al., 2024). This success positions the design of the underlying policy-optimization objective as a decisive determinant of how effectively verifiable reward signals translate into improved reasoning.

The GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed to operationalize RLVR at scale, and although its strong empirical performance has been validated across numerous studies, recent work such as DAPO (Yu et al., 2025) has exposed two critical and mutually reinforcing limitations that fundamentally constrain it. First, GRPO is prone to entropy collapse: as advantage values decay toward zero during the later stages of training, the learning signal is progressively silenced and optimization stalls. Second,

its indiscriminate weight-clipping discards rarely sampled tokens—precisely the signals that, by virtue of their low probability, carry the most information—thereby curtailing the diversity of model exploration. Far from being independent artifacts, these two failure modes share a common origin in the step-local, fixed-bound design of the underlying objective, which suggests that both can be addressed only by reconsidering that design itself.

Existing remedies address these symptoms only partially. DAPO, for instance, introduces the Clip-Higher strategy to counter entropy collapse by relaxing the upper clipping bound, together with a Dynamic Sampling strategy that filters out prompts with uniform rewards so that every batch retains only samples carrying valid gradients. These interventions, while beneficial, incur two notable and persistent costs. (i) **Response-Level Inefficiency**: because dynamic sampling discards and regenerates responses, it intrinsically lowers sampling efficiency and slows training convergence. (ii) **Token Clipping Deficit**: our analysis establishes that the optimal clipping bound is not constant but varies from token to token as a function of the token’s prior probability, so that any uniform upper bound is inherently suboptimal. Even approaches that avoid discarding zero-gradient responses outright leave both limitations fundamentally unresolved. Taken together, these observations motivate a method that treats the clipping bound and the advantage estimate not as fixed hyperparameters but as jointly adaptive quantities determined by the token- and step-level statistics of training.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a unified reinforcement-learning method that resolves both limitations at their source through three coordinated design choices, which together constitute the core contributions of this work. First, we develop a **smooth advantage standardization technique that standardizes advantages across cumulative steps**, directly eliminating the vanishing-gradient problem that arises when responses sharing an identical reward are assigned zero advantage. By aggregating the reward distribution of all cumulatively generated responses with that of the current step, the technique restores useful learning signals to otherwise discarded responses and thereby enables markedly more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping mechanism that adaptively adjusts the clipping bounds according to the old token-specific probabilities**, superseding the fixed—whether symmetric or asymmetric—clipping bounds of prior methods: tokens assigned lower prior probability are granted correspondingly wider clipping bounds, opening additional room for token-level exploration exactly where the high-entropy signal is most informative (see Section 2 and Appendix A.6). Third, DCPO computes the mean loss over the tokens within each individual response rather than over an entire batch; this Only-Token-Mean formulation preserves the relative advantage structure among responses to the same prompt and eliminates the dilution that batch-level averaging otherwise induces. Crucially, the three mechanisms are complementary by design rather than incidental: adaptive clipping restores token-level exploration, cumulative standardization restores response-level utilization, and the response-normalized loss preserves the advantage structure on which the other two depend. Their integration into a single coherent objective is what allows DCPO to convert generated data into usable learning signal far more efficiently than prior methods.

We evaluate these mechanisms on a training corpus that merges Math-DAPO-17k (Yu et al., 2025) with levels 3–5 of the MATH dataset (Liu et al., 2025), assessing performance on four widely used mathematical reasoning benchmarks. Across four distinct models spanning both instruct and base variants, and against GRPO, DAPO, and—where available—GSPO as baselines, DCPO consistently attains the strongest results, with the largest margins concentrated on the most challenging AIME benchmarks: on AIME24-Avg@32 under 32-fold sampling, DCPO-7B scores 38.8 against 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while on AIME25-Avg@32, DCPO-14B reaches 19.0 against 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). Critically, these accuracy gains are inseparable from the efficiency gains that produce them: DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude relative to the baselines, delivering superior end-task performance and substantially more efficient use of generated data in a single method.

For preliminary background, including the loss and advantage formulations of prior algorithms such as GRPO, DAPO, and GSPO, we refer the reader to Appendix A.2, against which the design choices introduced below can be read directly.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a standard mechanism in reinforcement learning for stabilizing policy updates and forestalling the large, destabilizing parameter shifts that off-policy optimization can induce. Its role becomes especially critical under importance sampling, the mechanism through which off-policy correction is realized. When importance sampling reweights samples by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, it yields an unbiased estimate of the expectation $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$; this unbiasedness, however, is purchased at the cost of potentially severe variance inflation, as characterized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To curb this variance inflation, prior works (e.g., Schulman et al. (2017)) constrain the probability ratio to a fixed clipping bound governed by the single hyperparameter ϵ , as stated in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, systematically constrains the model’s exploration of the diversity space, and does so most acutely in the very regions where the old policy assigns low probability mass. In those regions the absolute admissible bound on decreasing probabilities is inherently narrow, which suppresses the capacity for meaningful policy updates precisely where informative, low-probability tokens reside. The asymmetric clipping bounds adopted by DAPO relax this constraint only partially and leave the underlying limitation intact, since a bound that is fixed—symmetric or not—cannot adapt to the probability of the token it constrains. GSPO fares no better: by discarding entire responses whenever the sequence-level importance ratio exceeds its asymmetric bounds, it simultaneously wastes informative tokens and retains high-variance ones—a combination that destabilizes training rather than stabilizing it. The common shortcoming of all three schemes is thus a bound that is insensitive to token probability, which motivates making the admissible interval an explicit function of that probability.

Controlling the bias–variance trade-off therefore calls for a more principled alternative: one that makes the admissible interval depend explicitly on the token probability rather than holding it fixed, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting the identity $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero yields a closed-form admissible interval for $r(x)$. The result is a family of **dynamic-adaptive clipping bounds** that vary with the old probability $q(x)$ and are parameterized by the two hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping scheme systematically enlarges the admissible policy-update interval in low-probability regions, thereby restoring the exploration capacity that fixed bounds suppress, while leaving the tight interval for high-confidence tokens exactly unchanged. In this sense it strictly dominates fixed clipping: it coincides with the fixed bound wherever clipping is benign and relaxes it only where exploration is most needed. To guard against the instabilities that arbitrarily wide bounds could introduce, and following the dual-clipping method (Ye et al., 2020), we cap the maximum ratio at 10 for both positive and negative advantages, which prevents excessively large importance weights from overtraining the policy. The complete derivation of these bounds from the importance-sampling framework, together with a detailed analysis of their behavior, is provided in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at the i -th step, as given in Equation (11) of Appendix A.3. This step-local standardization induces two distinct and consequential failure modes. First, whenever sampling randomness drives all rewards at a given step to a common value, the standardized advantage collapses identically to zero, and the prompt is excluded from parameter updates even though its responses may encode valuable differences in reasoning trajectories. Second, the randomness inherent in high-entropy sampling can yield sharply skewed label counts, producing large step-to-step fluctuations in the standardized advantage—including outright sign reversals—that destabilize training. Both failure modes stem from estimating the advantage from a single step’s rewards in isolation, and both are therefore addressable by broadening the statistical base over which standardization is performed.

Recent evidence (Yue et al., 2025) establishes that RLVR refines a policy principally by adjusting the likelihood of correct samples rather than by fundamentally reshaping the underlying distribution. This finding licenses a key modeling choice that anchors our approach: the reward distribution of responses to a given prompt can be treated as draws from a single global distribution that persists throughout training. That premise directly motivates a *cumulative* standardization, defined in Equation (5), which estimates the advantage against the pooled statistics of all responses generated for the prompt rather than against those of the current step alone.

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics μ_{total}^i and σ_{total}^i are computed over all responses generated so far for the same prompt.

To temper the residual fluctuations of both the step-specific standardization ($\hat{A}_{new,j}^i$) and the cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each defined as a convex combination of the two standardization schemes whose step-dependent weights evolve monotonically with the training step i , as given in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As training progresses, $\hat{S}A_{new,j}^i$ shifts weight toward the current distribution, encoding a preference for adapting to the latest policy outputs, whereas $\hat{S}A_{total,j}^i$ shifts weight toward the cumulative distribution, treating the current step’s deviation as a fine-grained adjustment within the broader distribution. The final advantage $\hat{A}_j^i$ then selects, between these two smoothed estimates, the one of smaller absolute magnitude, thereby attenuating whichever source of fluctuation—cumulative or step-specific—momentarily dominates and, in doing so, systematically preserving training stability, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

A direct and decisive consequence of this construction is that, once a prompt has entered optimization, every response it generates in subsequent steps continues to contribute to the update irrespective of whether its step-local advantage vanishes. In particular, when identical rewards occur within a step, the affected responses still update the model through the residual advantage $\frac{1}{i} \hat{A}_{total,j}^i$, so that useful learning signals are retained rather than discarded. This is precisely the mechanism by which SAS converts the zero-gradient responses that GRPO wastes into an active learning signal, and it is the source of the substantial response-level efficiency gains reported in Section 5.3.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss in Equation (13), used by DAPO) equalizes the contribution of every token, but precisely in doing so it distorts the relative relationship between responses of differing lengths. A concrete instance makes the distortion explicit. Consider two responses to the same prompt: response A with advantage 1 and length 500, and response B with advantage 0.5 and length 1500. Under TLM, A is weighted by $\frac{500}{500+1500} = 0.25$ and B by $\frac{1500}{500+1500} = 0.75$, so that their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response B thus exerts the greater influence (0.375 versus 0.25)—a length-driven inversion of the intended advantage ordering, and one that grows more severe as response lengths diverge.

The **Sequence-Level Mean loss (SLM)** (the loss in Equation (12), used by GRPO) exhibits a complementary defect: by averaging across the batch it attenuates the magnitude of each advantage, effectively scaling it down by a factor of G , so that as the batch size grows this attenuation progressively erodes the correlations that the original advantage standardization was designed to encode. Since standardized advantages already carry well-defined relative relationships across responses, averaging over the $|o_i|$ tokens within a single response both preserves those relationships and distributes importance uniformly across that response’s tokens. We therefore average only over the tokens within each response and deliberately omit the batch-level average across responses—a formulation we term the **Only Token Mean loss (OTM)**, given in Equation (8). OTM thus avoids both the length-driven inversion of TLM and the advantage attenuation of SLM, retaining the intended advantage ordering exactly.

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO loss omits the KL-divergence constraint, and its lower and upper clipping bounds ε_{low} and $\varepsilon_{\text{high}}$ are instantiated directly by the dynamic-adaptive bounds derived in Section 2, while the advantage $\hat{A}_{i,t}$ is supplied by the smooth cumulative standardization of Equation (7). In this way all three mechanisms—OTM, SAS, and DAC—operate within a single, coherent objective rather than as separable add-ons.

4 EXPERIMENTAL SETUP

Models and Datasets To probe generality across both model scale and model type, we experiment with four models that jointly span instruct and base variants as well as general-purpose and math-specialized pretraining: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). This selection deliberately varies scale (1.5B to 14B), pretraining regime (general-purpose versus math-specialized), and post-training status (instruct versus base) so that any consistent advantage cannot be attributed to a single model family. Training draws on a merged corpus of approximately 25k mathematical problems that combines the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3–5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem is formatted with the Qwen-Math template (see Appendix A.8) for both training and evaluation, which enforces a consistent prompting protocol throughout and isolates the optimization method as the sole source of variation across all comparisons.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks that together span a broad difficulty spectrum, from grade-school through competition level: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, we report Avg@1 alone, since its larger size (500 problems) already yields a stable estimate. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32, so that greedy and sampling-based behavior are each assessed under a robust and comprehensive protocol that separates single-shot accuracy from distributional reliability.

- **Avg@1**: the standard accuracy under greedy decoding, which quantifies the quality of the model’s single most probable prediction.
- **Avg@32**: the mean accuracy over 32 responses sampled per problem at temperature 1.0 and top_p 1.0, which quantifies the robustness and stability of the trained policy distribution under stochastic decoding and thus measures the breadth of correct reasoning trajectories the policy can realize.

Baseline Settings We adopt GRPO and DAPO as baselines throughout, and additionally include the more recent GSPO on Qwen2.5-Math-7B and Qwen2.5-14B. The remaining configuration—comprising the per-method clipping thresholds ϵ_* , the codebase, and the common training hyperparameters—is specified in full in Appendix A.9. Every method is trained under identical data, prompting, and hyperparameter budgets, so that the resulting comparisons are strictly controlled and any performance difference is attributable to the optimization objective alone.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 reports the performance of GRPO, DAPO, GSPO, and our proposed DCPO on the four mathematical reasoning benchmarks—MATH500, AMC23, AIME24, and AIME25—under both greedy decoding (Avg@1) and sampling-based decoding (Avg@32), which jointly characterize each method’s accuracy, robustness, and generalization. The pattern is unambiguous: across every model scale from 1.5B to 14B, and across both instruct and base variants, DCPO attains the strongest average performance, matching or surpassing GRPO, DAPO, and—where available—GSPO in every configuration. Because this advantage holds uniformly across model capacities, pretraining regimes, and decoding conditions, it establishes DCPO as a general-purpose policy-optimization method for reinforcement learning of LLMs rather than an approach tuned to a particular benchmark or model family.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding both GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), surpassing GRPO (+1.3/+1.7) and remaining competitive with DAPO, improving Avg@1 by +1.0 while trailing by only a marginal -0.4 on Avg@32. On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), outperforming GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9) simultaneously. On the largest model tested, Qwen2.5-14B, DCPO delivers (53.2/39.0), with substantial gains over GRPO (+6.5/+7.7) and consistent advantages over DAPO (+1.3/+0.1) and GSPO (+4.2/+5.5) on both Avg@1 and Avg@32. Taken together, and with the single marginal exception noted above, these results demonstrate that DCPO improves reliably across the full range of model capacities examined and aligns the policy more effectively to the reward signal under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) DCPO’s most pronounced advantages emerge on AIME24 and AIME25, the high-difficulty competition benchmarks that most severely stress a model’s reasoning ability. Under Avg@32—which measures the capacity to produce correct responses under high-entropy sampling—DCPO exhibits markedly stronger robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, ahead of GRPO (32.1), DAPO (31.6), and GSPO (34.9), and on AIME25-Avg@32, DCPO-14B reaches 19.0, ahead of GRPO (10.5), DAPO (15.3), and GSPO (9.9). Since the sampling metric rewards the diversity and reliability of correct generations rather than a single greedy output, these margins establish that DCPO improves not merely average-case accuracy but also the breadth of correct reasoning trajectories the policy can realize. This joint optimization of reward fidelity and exploration under high-entropy decoding is exactly the property demanded by settings in which many plausible outputs are sampled per problem, and it is the behavioral counterpart of the efficiency gains analyzed in the remainder of this section.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of tokens that clipping excludes from the policy-gradient update during back-propagation; a lower *TCR* therefore corresponds to a larger fraction of the generated sequence contributing to learning, and hence to more efficient use of each sampled response. As Figure 1 shows, the *TCR* behaves in sharply different ways across the reinforcement-learning methods considered, and these differences track the exploration behavior that

Table 1: Performance across the four mathematical reasoning benchmarks (MATH500, AMC23, AIME24, AIME25), reported per model. Avg@1 is measured under greedy decoding and Avg@32 is averaged over 32 responses sampled at temperature 1.0 and top-p 1.0. Boldface marks the best value in each column, and Δ_{baseline} reports DCPO’s margin over the corresponding baseline, with red denoting an improvement. Across all four models and both decoding regimes, DCPO attains the strongest **Average** score, and the advantage over every baseline widens on the hardest AIME benchmarks under sampling.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

the clipping design is intended to control.

$$TCR = \text{Average} \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } \text{micro}_m}{\text{Total number of tokens in } \text{micro}_m} \right) \quad (9)$$

where micro_m denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO, the TCR follows model-size-dependent trajectories: it rises with training steps for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) yet gradually declines for the larger ones (Qwen2.5-Math-7B and Qwen2.5-14B). DAPO, by contrast, exhibits a monotonically increasing TCR across all models, so that as training proceeds an ever-larger share of policy updates is computed from partial or truncated responses. Both GRPO and DAPO, moreover, display pronounced fluctuations and occasional anomalous spikes in the TCR , which inject instability into optimization and can precipitate entropy collapse. GSPO is more wasteful still: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B—far above that of the token-level clipping methods—so that it discards

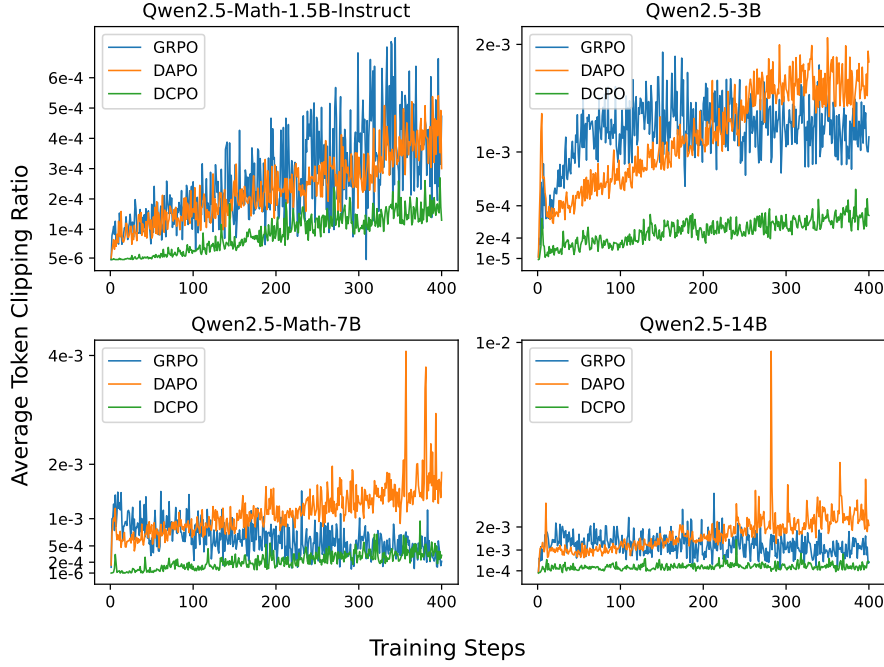


Figure 1: Token Clipping Ratio (TCR) over training for each method across the four models. Throughout training and independent of model size, DCPO sustains a TCR that is substantially lower—by roughly an order of magnitude—and markedly more stable than that of both GRPO and DAPO, indicating that far fewer tokens are excluded from the policy-gradient update.

a correspondingly larger volume of informative tokens, a direct consequence of its sequence-level clipping criterion.

DCPO, in stark contrast, sustains a stable and efficient training regime. Independent of model size or training stage, its TCR remains nearly constant and stays an order of magnitude below that of GRPO and DAPO. This uniformly low and stable TCR entails that DCPO discards far fewer tokens, allowing a greater share of complete response sequences to inform each policy update and thereby opening additional room for the model to explore rare-token diversity. The net effect is a clear and consistent improvement in sample efficiency over all baselines, obtained while keeping outlier variation tightly bounded rather than trading stability for efficiency.

Taking Qwen2.5-Math-7B as a representative case, the token distribution concentrates rapidly on high-confidence tokens: after roughly 60 steps about 95% of generated tokens satisfy $q(x) > 0.9$, a fraction that rises to about 97% by roughly 100 steps and continues climbing thereafter. As illustrated in Figure 4d, once the old probability satisfies $q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$, the new probability $p(x)$ contributes to the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never discarded under any clipping method. The overwhelming majority of tokens therefore reside in a region that is inherently immune to clipping, which accounts for the low overall TCR observed in practice. The consequential clipping is thus confined to the low-probability tail. Consistent with this, Figure 1 shows the TCR rising during the later phases of most baseline runs, signaling that a growing proportion of the remaining low-probability tokens is being discarded—precisely the tokens whose loss most severely restricts the model’s ability to explore rare-token diversity.

Prior work (Wang et al., 2025) establishes that **high-entropy tokens**—which are exactly the low-probability tokens—are the primary drivers of the emergence of reasoning capabilities in LLMs. Because entropy and probability are inversely related, rarer tokens carry proportionally more information; granting such tokens a wider admissible update range therefore enlarges the model’s effective exploration space and directly facilitates the acquisition of diverse reasoning behaviors. It follows that a clipping rule which preferentially spares these tokens, rather than discarding them uniformly, is exactly what an effective RLVR objective requires.

These empirical and theoretical observations jointly and directly motivate our **dynamic-adaptive clipping** scheme, which relaxes the clipping constraint for tokens with $q(x) < \frac{1}{1+\epsilon}$ while retaining the safe interval for high-confidence tokens. By adaptively widening the update interval for the high-entropy tail of the distribution, the scheme restores the model’s capacity to learn from rare, informative tokens and, in consequence, strengthens its reasoning ability. The low and stable *TCR* in Figure 1 is the empirical signature of this mechanism, and it links the design of DAC to the end-task gains reported in Table 1.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a response contributes to a policy update only when it carries a nonzero gradient. We quantify how effectively this condition is met through the **Response Utilization Ratio** (*RUR*), defined as the percentage of generated responses with nonzero advantage that actually participate in the policy update; a higher *RUR* thus signifies more efficient exploitation of the generated data and, correspondingly, less wasted sampling and compute.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

As Table 2 summarizes, the average *RUR* under GRPO is strikingly low—only 44.6% averaged across the four models—so that more than half of all generated responses are discarded, a substantial inefficiency in the exploitation of exploratory diversity. DCPO, by contrast, sustains an average *RUR* of roughly 70% after the first epoch, an absolute gain of 28% and a relative improvement of 64% over GRPO, and it does so consistently across all four models rather than only in favorable configurations.

Table 2: Average *RUR* over 400 training steps, reported per model. DCPO raises response utilization to roughly 70% on every model, well above both GRPO (43.8% on average) and GSPO (“-” denotes settings in which GSPO was not evaluated), demonstrating that the gain is uniform across model scale and pretraining regime.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* degrades sharply over the course of training, independent of model size or type, falling from above 90% to below 50%; on Qwen2.5-Math-7B it collapses to as low as 30% after roughly 200 steps. This decline is the direct signature of GRPO’s step-local advantage standardization, which generates a large and growing fraction of zero gradients and thereby renders exploration increasingly inefficient precisely in the later stages of training, when informative gradients are most scarce. DCPO, driven by smooth advantage standardization, instead maintains a consistently high *RUR*: after the first epoch it stabilizes around 70% and continues to rise as training proceeds, confirming that the mechanism arrests rather than merely delays the utilization collapse observed under GRPO.

The *RUR* exceeds 90% during the first few steps but, from the second epoch onward, settles around 70% rather than reaching 100%. Examining precisely why it settles there, rather than dismissing the remaining gap as residual inefficiency, both clarifies the behavior of the SAS mechanism and shows that the plateau reflects a deliberate and desirable property of the method.

First, the initial *RUR* above 90% is largely an artifact of early-stage outputs that violate the required formatting, incur negative rewards, and thereby inflate the metric. That these outputs are quickly corrected indicates that the models rapidly internalize the fixed answer patterns, which is precisely why subsequent training should redirect emphasis toward exploring genuine reasoning capabilities rather than format compliance. Second, our analysis identifies two principal sources of persistently zero advantage that hold the *RUR* near 70% rather than at 100%. (i) Nearly half of the problems are easy enough that the model solves them consistently; excluding these from updates is not a loss but a justified economy, since the model’s core capabilities require no further refinement on them. (ii) The remaining consistently zero-advantage cases correspond to persistent errors, which arise for two distinct reasons. On the one hand, some problems remain too difficult for the current model even given a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset contains items

whose labels were converted to mislabeled integers, introducing spurious “incorrect” targets. In both cases, discarding zero-advantage responses functions as an implicit data filter, mitigating the risk of updating the model on noisy or mislabeled data. The plateau near 70% is therefore a designed feature of the mechanism rather than a shortfall of it, and attempting to force utilization toward 100% would reintroduce exactly the noise the plateau screens out.

Because DAPO, unlike GRPO and DCPO, discards responses with identical labels from the generated data, a direct *RUR* comparison is not meaningful; we therefore gauge DAPO’s efficiency through the number of generated responses and the wall-clock time required to complete an equal number of update steps. To offset the low sampling efficiency that GRPO’s low *RUR* would otherwise cause, we set DAPO’s generation batch size to 1–4 times the training batch size and adjust it dynamically. Even under this favorable configuration, across the four models DAPO requires 3 to 5 times more generated responses than DCPO to reach the same number of parameter updates, which at least doubles the training GPU hours while yielding no accompanying performance gain. This inefficiency compounds with scale: on Qwen2.5-14B, the fraction of responses DAPO actually uses falls below 30% after about 300 steps. Dynamic sampling therefore combines low data utilization with poor training efficiency and forfeits substantial opportunity to explore model diversity—precisely the trade-off that DCPO’s cumulative standardization is designed to avoid.

In summary, DCPO advances decisively beyond both GRPO and DAPO along a single, coherent axis: it substantially raises response utilization and training efficiency, converts a far larger share of the sampled response pool into usable exploration signal for model diversity, and translates that efficiency directly into the improved end-task performance documented in Table 1.

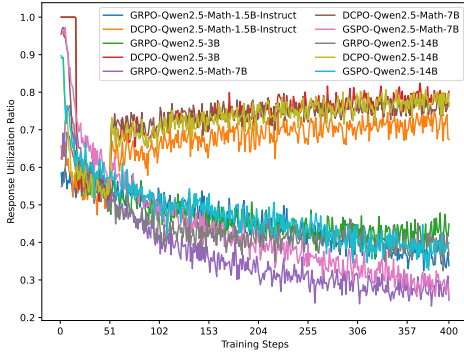


Figure 2: Evolution of the *RUR* during training. GRPO’s utilization decays sharply toward and below 50%, whereas DCPO stabilizes near 70% and remains there—continuing to rise—for the remainder of training.

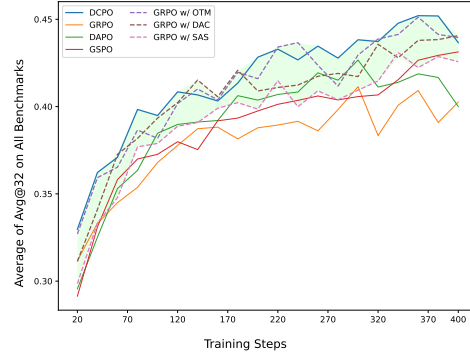


Figure 3: Component ablation on Qwen2.5-Math-7B, measured by average Avg@32 over training. Each DCPO component (OTM, SAS, DAC) is added individually to GRPO, and their combination constitutes the full DCPO; every single component surpasses GRPO, and their integration into the full method dominates all baselines, with the margin widening in the later stages of training.

5.4 ABLATION STUDIES

To isolate the contribution of each DCPO component, we conduct a controlled ablation on Qwen2.5-Math-7B using **Avg@32**, the metric most sensitive to the robustness and stability of the learned policy distribution. Each variant modifies exactly one component of the baseline GRPO framework while holding all other settings fixed and removing the KL-divergence term, so that the configuration is aligned with DAPO, GSPO, and the full DCPO; this design attributes each observed gain unambiguously to the component it isolates. The corresponding learning curves from step 20 to step 400 are reported in Figure 3.

Each of the three components, added in isolation, drives a monotone improvement in reasoning ability: every variant surpasses GRPO and matches or exceeds DAPO and GSPO. Their combination then produces a substantial aggregate gain over all baselines, confirming the effectiveness of the complete DCPO pipeline for reinforcement-learning tasks. The distinction sharpens in the later stages of training, where the baselines plateau while DCPO continues to trend upward, both at the level of

individual components and in aggregate—evidence that the mechanisms address a limitation that intensifies over training rather than a transient one. The ablation thus establishes that each component contributes positively to training efficiency and that their integration yields cumulative gains together with greater stability. We detail each contribution below.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each individual response rather than across the whole batch already lifts GRPO above the baselines, which suffer from lower data utilization and more volatile performance, confirming that preserving the response-level advantage structure is itself a source of gain. (ii) **Smooth Advantage Standardization (SAS)**: replacing the stepwise advantage with the cumulative smoothed advantage markedly improves GRPO and approaches DAPO—direct evidence that aggregating reward statistics across training steps stabilizes optimization and raises final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting a probability-adaptive clipping boundary for GRPO’s fixed boundary yields the largest single-component gain—DAC alone surpasses GRPO, DAPO, and GSPO, and does so more stably than OTM, corroborating that token-level exploration is the dominant lever among the three. (iv) **Full DCPO (OTM + SAS + DAC)**: integrating the three mechanisms delivers the best and most stable results overall, decisively outperforming GRPO, DAPO, and GSPO. Because the combined method exceeds every single-component variant, the three design choices are shown to contribute distinct and complementary benefits to data-efficient, stable reinforcement learning, jointly improving both performance and robustness for large language models.

Overall, the ablation confirms that every DCPO component contributes positively to end performance and that their combination produces substantial cumulative gains, establishing the proposed mechanisms as effective and general levers for improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

This work resolves two central limitations of existing policy-optimization methods: the restricted token-level exploration imposed by fixed clipping in low-probability regions, and the low sample efficiency induced by step-local reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement-learning pipeline that strengthens the reasoning capabilities of large language models through two complementary innovations: (i) a **dynamically adaptive token-level clipping** mechanism that widens the update interval as a monotone function of the old-policy probability, thereby enabling efficient exploration of rare, low-probability tokens, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates reward statistics across optimization steps, thereby damping the reward randomness introduced by high-entropy sampling. Extensive experiments on four mathematical reasoning benchmarks across four models—spanning instruct and base variants at scales from 1.5B to 14B—substantiate these claims: DCPO-7B, for instance, attains an Avg@32 of 38.8 on the challenging AIME24 benchmark, surpassing both GRPO (32.1) and DAPO (31.6). Beyond accuracy, and as the shared source of it, DCPO raises the response utilization ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. Together these results establish DCPO as a general and effective method that markedly improves data utilization and diversity exploration in RLVR, with consistent gains across model scale and pretraining regime. Building on this foundation, future work will extend DCPO to further domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_\theta(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_\theta(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_\theta \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is obtained by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards over the G responses generated at the i -th step alone, as given in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for a given prompt are identical, the corresponding responses receive zero advantage and contribute nothing to the update, so the sampled data are wasted—the failure mode that the cumulative standardization of Appendix A.7 is designed to eliminate.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R via a rule-based reward function, and estimates token-level advantages as in Equation (11). The model parameters are then updated according to Equation (12), the **Sequence-Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch; the $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against excessive deviation from a reference policy (e.g., a pre-trained model); and the fixed clipping threshold ε bounds the change in token probability ratios to prevent overly large updates. Widely adopted though it is, this batch-level averaging inherently dilutes per-response importance by the factor G , which is precisely the effect that the OTM formulation removes.

A.3.2 DAPO

Building on GRPO, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**. Observing that longer sequences exert a disproportionate influence on the GRPO gradient, they normalize by the total token count across the entire batch, yielding the **Token-Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards every prompt whose responses all share an identical reward and regenerates responses to restore the batch size—a procedure that restores valid gradients but at the cost of severe data inefficiency, since the discarded responses are re-sampled rather than reused.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into gradient estimates—noise that accumulates over long sequences and is further amplified by the clipping operation itself. GSPO mitigates this by replacing token-wise clipping with sequence-wise clipping, thereby producing a more stable training signal at the sequence level, albeit at a coarser granularity.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_\theta(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)} \right) \right)$$

This sequence-level criterion, however, discards more than 10% of responses that carry nonzero advantages, so its stability is purchased at the price of considerable data waste and reduced training efficiency. Moreover, while GRPO discards responses exhibiting sequence-level high variance, it still retains a large number of responses with token-level high entropy, and these continue to destabilize training—so that the sequence-level remedy addresses only one of the two variance sources.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward model that scores each generated response jointly on answer correctness and format compliance, as defined in Equation (15). Concretely, a reward of +1 is assigned when both the format and the answer are correct, 0 when the format is correct but the answer is incorrect, and −1 when the format itself is incorrect, so that the reward simultaneously enforces well-formed outputs and rewards correct reasoning.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response within the group for a given problem at the i -th training step, and the reward is determined jointly by the format compliance and answer correctness of that response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning for stabilizing policy optimization and preventing large, destabilizing parameter updates. Its importance is heightened whenever importance sampling is used to estimate expected values under a new policy from samples drawn by an old one, since it is precisely in this off-policy setting that unbounded ratios can inflate the estimator variance without bound. This appendix derives, from that variance analysis, the dynamic-adaptive bounds summarized in Section 2.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be re-expressed exactly as an expectation under the old probability $q(x)$ through the importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be substantially inflated—a well-known difficulty of importance sampling that motivates the clipping constraint. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator computed under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

Taking the difference between the importance-sampled variance and the true variance makes the potential for inflation explicit, and identifies the ratio $\frac{p(x)}{q(x)}$ as the quantity that must be controlled, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To contain this discrepancy, many prior works—including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently constrains the policy’s capacity for exploration, and this limitation is most damaging precisely where the old policy assigns low probability mass. In such regions the absolute bound on low probabilities is severely restricted, which suppresses the potential for meaningful policy updates. Asymmetric clipping bounds, as introduced by DAPO, address only

the symmetry of the constraint and leave the fundamental limitation of a non-adaptive bound fully intact.

Moreover, recent studies such as Yue et al. (2025) indicate that the reasoning paths exploited by RLVR models already reside within the sampling distribution of the base model—a conclusion reinforced by the minimal parameter shift observed during off-policy batch updates, which manifests as a small fraction of clipped parameters. Motivated by this stability, we derive a more principled alternative that constrains the probability ratio $r(x)$ through a dynamic-adaptive mechanism, incorporating the probability directly into the restriction as in Equation (21) and thereby superseding the conventional fixed clipping of Equation (20).

$$|(\frac{p(x)}{q(x)} - 1)p(x)| \leq \epsilon \quad (21)$$

Substituting the identity $p(x) = r(x)q(x)$ into the preceding inequality and expanding yields the equivalent expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, it must respect the fundamental constraints on probability distributions and their ratios. As stated in Equation (23), the probabilities $p(x)$ and $q(x)$ both lie in $[0, 1]$; consequently the probability ratio $r(x)$ is non-negative, as is the clipping hyperparameter ϵ . These constraints delimit the admissible region within which the bounds below must be resolved.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To guarantee both the validity and the practical applicability of the clipping bound derived from Equation (22), we specify the necessary conditions on its bounds, as set out in Equation (24), distinguishing the four boundary functions by whether they bound $r(x)$ from above or below.

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24), the probability ratio $r(x)$ must satisfy the following case-dependent conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we restrict attention to the overlapping region centered on the ratio value 1, since this region both minimizes the discrepancy between the two conditions and preserves consistency with the conventional fixed bound of GRPO; the resulting validity conditions are:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we guarantee that the radicand remains non-negative by taking its maximum with zero, e.g. $\max(1 - \frac{4\epsilon}{q(x)}, 0)$, which keeps the bounds well defined across the entire probability range. This finally yields the conditions below, which constitute the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we assign distinct thresholds ϵ_{high} and ϵ_{low} so that the dynamic-adaptive bound coincides exactly with the GRPO bound at the designated reference points, guaranteeing backward compatibility with fixed clipping in the high-confidence regime.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To make our dynamic-adaptive clipping bounds coincide with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we determine the specific values of ϵ_{low} and ϵ_{high} by solving the system of equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

Under the standard practice of prior work (e.g., GRPO), where the fixed clipping threshold is $\epsilon = 0.2$, solving these equations gives our hyperparameters as approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although $\epsilon_{low} \neq \epsilon_{high}$, this asymmetry serves a fundamentally different purpose from the asymmetric fixed clipping of DAPO: rather than statically relaxing the upper limit, it calibrates a probability-dependent bound so as to align exactly with the fixed bound at the reference points, so the asymmetry is a consequence of the calibration rather than a free design choice.

Dynamic versus fixed clipping: Figure 4 contrasts the behavior of the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) with that of our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed scheme supplies a constant relative width 2ϵ , so that as $q(x)$ decreases the absolute exploration space it permits shrinks linearly (Figure 4a). The dynamic scheme, by contrast, widens its relative interval inversely with $q(x)$, so that the absolute admissible exploration space remains approximately constant across the entire probability range—the property that equalizes exploration across common and rare tokens alike.

Effect of decreasing $q(x)$: As $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains independent of $q(x)$, so the model gains no additional room to explore low-probability tokens; the dynamic-adaptive bound, in contrast, expands rapidly (Figure 4b), granting a progressively larger relative space to rare tokens—exactly the behavior required for efficient exploration in RL.

Behavior for extremely low probabilities: Figure 4c depicts the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound on $r(x)$ grows rapidly, at a rate inversely proportional to the square root of $q(x)$, whereas the fixed bound stays constant. To forestall the attendant instabilities (gradient vanishing, explosion, and the like), we follow the dual-clipping strategy (Ye et al., 2020) and impose a hard ceiling $r_{max} = 10$. The gray region in Figure 4c, corresponding to approximately $q(x) \leq 2 \times 10^{-3}$, marks where the adaptive upper limit is clamped to r_{max} , so that the unbounded growth is confined to a vanishingly small probability range.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$ the dynamic-adaptive scheme yields $dynamic_{low} = 0.5$, whereas the fixed scheme retains the constant $fixed_{low} = 1 - \epsilon = 0.8$; once $q(x)$ exceeds this threshold, the dynamic-adaptive lower bound contracts smoothly toward the fixed value. Symmetrically, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\epsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval

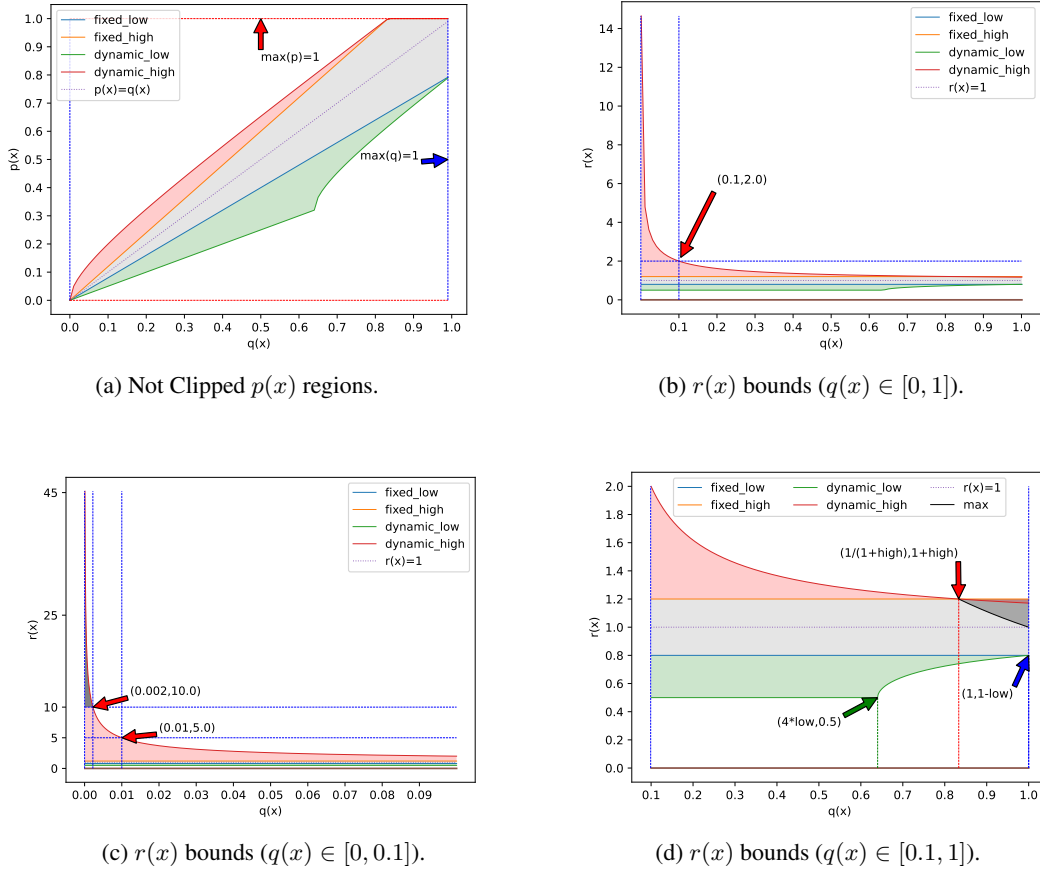


Figure 4: Comparison of clipping bounds as a function of the old probability $q(x)$: fixed clipping ($\epsilon = 0.2$) versus dynamic-adaptive clipping ($\epsilon_{\text{low}} = 0.16, \epsilon_{\text{high}} = 0.2$). As $q(x)$ decreases, the dynamic bound widens monotonically, expanding the admissible update interval for rare tokens, whereas for high-probability tokens it coincides with the fixed bound. The two schemes therefore agree exactly where clipping is benign and diverge precisely where exploration is most needed, which is the defining property of the proposed scheme.

$[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$ therefore delivers a probability-dependent widening of the admissible region for rare tokens while preserving the tightness of the fixed interval for common tokens (Figure 4d)—a widening that is thus targeted rather than indiscriminate.

Positive versus negative advantage: The loss in Equation (8) comprises both a positive- and a negative-advantage term. Under the minimum operation, the ratio $r(x)$ for a positive advantage is clipped to the interval $[0, dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$, whereas for a negative advantage it is clipped to $[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), \infty)$. Because an excessively large $r(x)$ can destabilize training, we cap the ratio at $r_{\text{max}} = 10$ in both cases; this safeguard, following Ye et al. (2020), guarantees that the adaptive limits never produce excessively high importance weights in either advantage regime.

In short, for both positive and negative advantages, dynamic-adaptive clipping affords the model strictly more room to explore diversity for low-probability rare tokens than fixed clipping does, while remaining indistinguishable from it for high-confidence tokens. It thereby dominates fixed clipping without introducing any new instability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i index the iteration at which prompts are used to generate responses and update the model parameters. We define below the reward statistics on which the cumulative and smooth advantage standardization depends.

The mean rewards of the responses are given by the family of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The corresponding variances of the response rewards are given by the family of functions in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource usage, we compute σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) through the equivalent recursions expressed in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than by direct accumulation over the original rewards. Because the two formulations are algebraically identical, the recursive form incurs no approximation while avoiding the storage of the full reward history, so cumulative standardization adds only $O(1)$ memory per prompt.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments in this paper employ the following prompting template, adapted from the Qwen-Math template (Yang et al., 2024b) and applied uniformly to both training and evaluation so that the prompt format is never a confounding variable.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

We configure the clipping thresholds per method as follows: a symmetric threshold $\epsilon = 0.2$ for GRPO; lower/upper thresholds (0.2, 0.28) for DAPO, augmented with a 512-token soft-punish cache and a maximum probability ratio of 10 for negative advantage; lower/upper thresholds ($3e-4$, $4e-4$) for GSPO; and low/high thresholds (0.16, 0.2) for DCPO. The DCPO thresholds are selected so as to preserve exactly the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$

as GRPO (Figure 4d), which ensures that DCPO introduces no additional tuning burden relative to GRPO; unlike DAPO, we cap the maximum probability ratio at 10 for both positive and negative advantages.

Training runs for 400 steps with a generation batch size of 512 and a parameter-update mini-batch size of 32. Using temperature 1 and top-p 1, we generate $G=16$ responses per problem, with the prompt and response maximum lengths set to 1,024 and 3,072 tokens, respectively—the latter far below the 20k maximum adopted in the original DAPO, which underscores that DCPO attains its gains under a markedly more economical generation budget.

Our training codebase is adapted from Verl (Sheng et al., 2024) and runs on $32 \times \text{H20}$ GPUs; unless stated otherwise, all hyperparameters match those of DAPO, so that the comparison isolates the optimization objective. For loss computation we retain the exact $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation of GRPO and DAPO rather than the approximate weighting used in Verl, which guarantees numerical correctness of the reported losses across differing microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

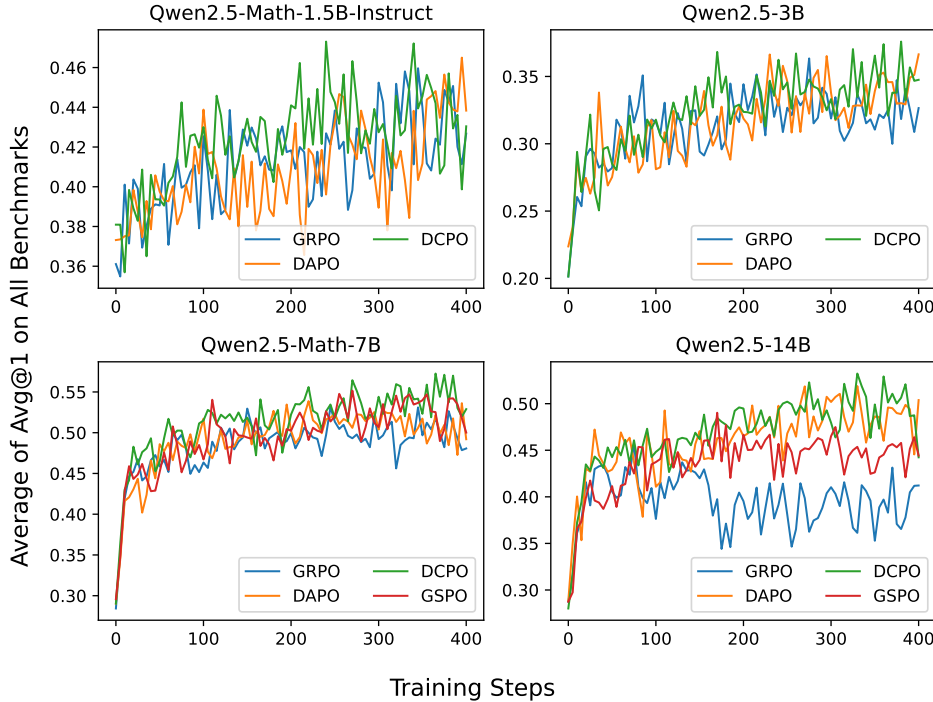


Figure 5: Avg@1 performance over training, averaged across the four benchmarks, for each model and method. DCPO consistently leads the baselines at every model scale while following a markedly smoother trajectory, free of the collapse seen in GRPO and the large fluctuations seen in DAPO.

Avg@1 Trend Figure 5 reports the average Avg@1 metric, obtained by averaging the greedy-decoding (Avg@1) results over the MATH500, AMC23, AIME24, and AIME25 benchmarks, which together characterize the model’s best-prediction performance. Across all four models, DCPO consistently outperforms both GRPO and DAPO, and it likewise surpasses GSPO on the two models where GSPO was evaluated. Its trajectory is, moreover, markedly more stable than those of the baselines: it is free of the pronounced performance drop exhibited by GRPO and of the large fluctuations exhibited by DAPO. These trends establish that DCPO elicits stronger reasoning capabilities over the course of training and, by avoiding late-stage collapse, leaves greater headroom for the model to attain still higher performance.

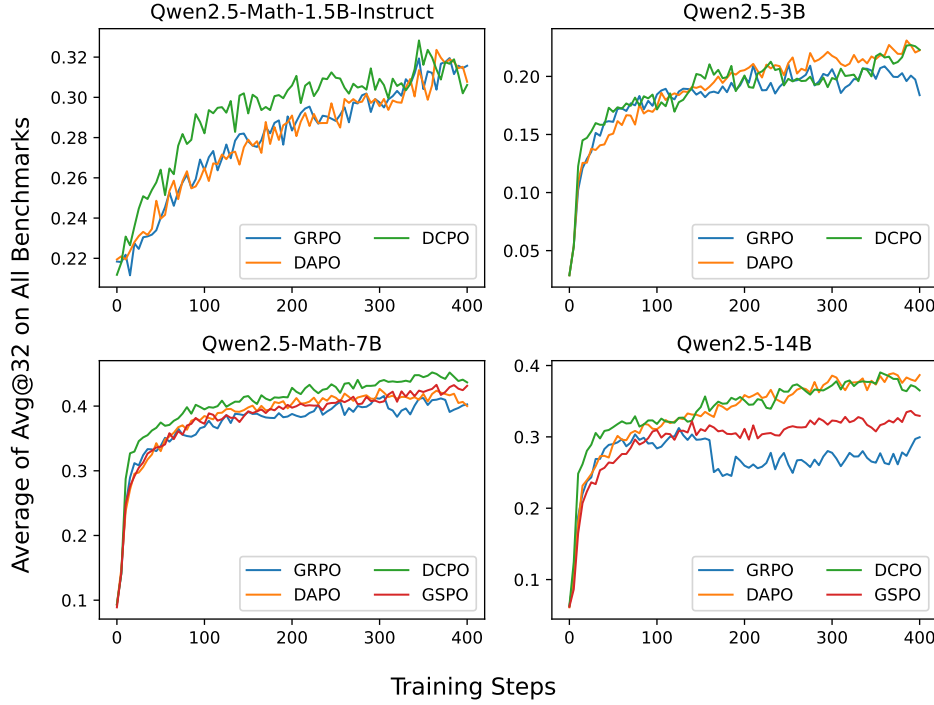


Figure 6: Avg@32 performance over training across the AMC23, AIME24, and AIME25 benchmarks, for each model and method. DCPO exhibits the fastest early-stage growth and sustains a stable upward trend under sampling across all model scales, in contrast to the fluctuation and occasional collapse of the baselines.

Avg@32 Trend Figure 6 compares the Avg@32 performance of DCPO against GRPO, DAPO, and—where available—GSPO across model sizes on the AMC23, AIME24, and AIME25 benchmarks, and it demonstrates DCPO’s exceptional robustness and stability under sampling. GRPO, by contrast, exhibits substantial fluctuation and occasional collapse: on Qwen2.5-14B it suffers a marked performance drop after approximately 170 steps. Across every evaluated model (1.5B to 14B), the DCPO trajectory trends consistently upward with minimal fluctuation, sustaining the fastest early-stage growth rate and preserving that upward trend thereafter. Relative to DAPO, DCPO furthermore attains superior performance while halving both training time and resource consumption—a simultaneous improvement in accuracy and cost rather than a trade-off between them. These results jointly establish the effectiveness and robustness of dynamic-adaptive clipping and smooth advantage standardization in improving diversity exploration at both the response and label levels.

Entropy Trend Figure 7 shows that, after an initial warm-up, GRPO’s entropy decays sharply and then settles into a flat, low-entropy regime. This entropy collapse reflects a rapid loss of policy randomness, which reduces the diversity of sampled responses and thereby constrains subsequent policy optimization, mirroring the utilization collapse documented in Section 5.3.

The behavior of DAPO and DCPO is model-dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the trends diverge markedly: DAPO sustains the highest entropy, trending clearly upward but with large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy closely tracks that of DCPO, and their average metric performance is correspondingly comparable; on Qwen2.5-14B, by contrast, GSPO’s entropy follows a fluctuating downward trend that yields poor results. DCPO instead settles at a moderate entropy level, tracing a substantially smoother curve confined to a bounded band between the low-entropy regime of GRPO and the high-variance regime of DAPO.

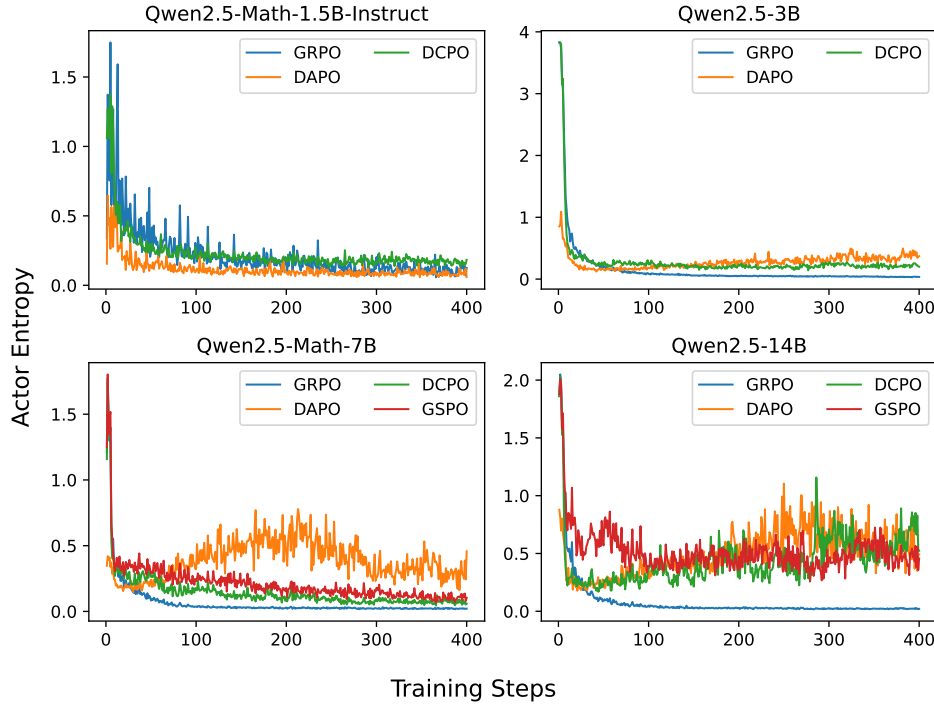


Figure 7: Policy entropy over training for each model and method. DCPO maintains entropy within a moderate, stable band, positioned between the low-entropy regime of GRPO and the high-variance regime of DAPO—the balance associated with its strongest and most stable end-task performance.

These observations point to a U-shaped relationship between entropy and training performance: excessively low entropy starves the learner of exploratory signal and induces premature convergence, whereas excessively high or unstable entropy undermines training stability. By holding entropy within a moderate, well-balanced range, DCPO reconciles these competing pressures, promoting convergence and sufficient exploration at once—a benefit that becomes more pronounced as model capacity increases. This entropy-regulating behavior is therefore a principal mechanism underlying DCPO’s superior reasoning performance and its more robust policy learning across models of varying sizes, and it ties the entropy dynamics directly to the accuracy and efficiency gains reported throughout the paper.